CHAPTER

27

Methylsulfonylmethane as an antioxidant and its use in pathology

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List of abbreviations

BBB	blood–brain barrier
CAT	catalase
CCl_4	carbon tetrachloride
DMSO	dimethylsulfoxide
GI	gastrointestinal
GRAS	generally recognized as safe
GPx	glutathione peroxidase
GR	glutathione reductase
GSH	reduced glutathione
GSSG	oxidized glutathione
GST	glutathione *S*-transferase
H_2O_2	hydrogen peroxide
HIV	human immunodeficiency virus
IBD	inflammatory bowel disease
IL	interleukin
iNOS	inducible nitric oxide synthase
LPS	lipopolysaccharide
MPO	myeloperoxidase
MSM	methylsulfonylmethane
$NADP^+$	oxidized nicotinamide adenine dinucleotide diphosphate
NADPH	reduced nicotinamide adenine dinucleotide diphosphate
NFκB	nuclear Factor Kappa-light-chain-enhancer of activated B cells
NLRP3	Nucleotide-binding domain, leucine-rich repeat family pyrin domain containing 3
NO	nitric oxide
NOAEL	no-observed-adverse-event-level
NOX	NADPH oxidase
Nrf2	nuclear factor (erythroid-derived 2)-like 2
O^{2-}	superoxide anion
OH^-	hydroxyl radical
$ONOO^-$	peroxynitrite radical
OPD	*O*-phenylenediamine
PUFA	polyunsaturated fatty acid
ROS	reactive oxygen species
SOD	superoxide dismutase
TAC	total antioxidant capacity
TEAC	Trolox equivalent antioxidant capacity
TLR	toll-like receptor
TNFα	tumor necrosis factor-α
Trx/Prx	thioredoxin/peroxiredoxin system

Introduction

Methylsulfonylmethane (MSM) is an organosulfur compound of the Earth's sulfur cycle along with dimethylsulfide and dimethylsulfoxide (DMSO). Beginning in the early 1960s, researchers began experimenting with DMSO as a cryopreservant for organ transplantation with much success. MSM later gained attention after Williams and colleagues reported that DMSO was oxidized to MSM in rabbits.[1] Combined with the fact that MSM did not have the malodorous side effects that DMSO possessed, MSM gained traction as the more commonly used supplement. Since these early investigations, the biological effects of MSM have expanded greatly with much research demonstrating the antiinflammatory, immunomodulatory, and antioxidant actions of MSM.

With these early investigations leading to the use of MSM as a dietary supplement came the sponsorship of multiple private industry toxicity reports. Table 27.1 displays MSM toxicity summaries from both sponsored toxicity reports and published in vivo and in vitro data. The data suggest that MSM is nontoxic at relatively high dosages, for example, the no-observed-adverse-event-level (NOAEL) is listed at 5 g/kg when given orally or intraperitoneally to mice or rats.[2] In fact, to our knowledge, only a single mortality has occurred in any MSM toxicity study using a dosage of 15,380 mg MSM/kg of body weight. This study reported the acute oral LD_{50} of $\geq$17,020 mg MSM/kg of body weight. That said, doses used in humans are much lower, with most doses being

Pathology.
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TABLE 27.1 MSM toxicity summary.

Cell line	Incubation time	Dosage (IC50)
Human smooth muscle	96 h	1% (reversible at 1%; partially reversible at 2%−3%; irreversible at 4%)
Human endothelial	96 h	1.8% (completely reversible up to 4%)
SK-BR3 (human breast cancer)	24 h	300 mM
MDA-MB231 (human breast adenocarcinoma)	24 h	300 mM
AGS (human gastric adenocarcinoma)	72 h	297.8 mM
HepG2 (human liver carcinoma)	72 h	232.3 mM
KYSE-30 (human esophageal carcinoma)	72 h	297.2 mM
YD-38 (human gingival carcinoma)	24 h	200 mM

Species	Route	Duration	NOAEL
Acute **≤ *15 days***			
Mice	Oral	*Not stated (acute)*	5 g/kg BW
Mice	Intraperitoneal	*Not stated (acute)*	5 g/kg BW
Mice	Oral gavage	14 days	2 g/kg BW
Mice	Oral gavage	15 days	5.0 g/kg BW
Rat	Intragastric	6 days	20 g/kg BW
Rat[a]	Intragastric	14 days	10.25 g/kg BW
			LD50 ≥ 17.02 g/kg BW
Rat	Intraperitoneal	*Not stated (acute)*	5 g/kg BW
Rat	Oral gavage	14 days	5.0 g/kg BW
Rat	Oral gavage	14 days	2 g/kg BW
Rat	Serial intranasal injections	*Not stated (∼3 days)*	0.6 ml/kg BW
Rat	Oral gavage	15 days	2 g/kg BW
Rabbit	Dermal	14 days	5.0 g/kg BW
Dog	Oral	*Not stated*	2 g/kg BW
Subacute			
Gestating Rat	Oral gavage (14 days)	21 days	1 g/kg BW/day
Cow	Oral	30 days	1.2 g/kg BW/day
Subchronic			
Rats	Oral gavage	90 days	1.5 g/kg BW/day
Mice	Oral	91 days	1.5 g/kg BW
Rat	Oral	90 days	1.5 g/kg BW
Horse	Oral syringe	84 days	0.04 g/kg BW/day

[a]*Study with reported mortality.*
In vitro and in vivo toxicity studies summarized from peer-reviewed studies and industry-sponsored reports.

between 1 and 6 g/day. Aided by these toxicity reports, Bergstrom Nutrition's OptiMSM was awarded the generally recognized as safe (GRAS) status by the United States Food and Drug Administration (USFDA).

Pharmacokinetic studies in rats using MSM with radiolabeled sulfur have yielded interesting results. In one such study, MSM was shown to be primarily eliminated in the urine while the remaining MSM was widely distributed throughout bodily tissues.[3] The following study elaborated on these findings by quantifying the amount of MSM reaching various organs.[4] An overview of the findings from these studies is displayed in Fig. 27.1. Work in humans indicate that plasma levels appear to stabilize after about 4 weeks of daily supplementation, with stable trough concentrations being dose-dependent.[5] The tissue distribution

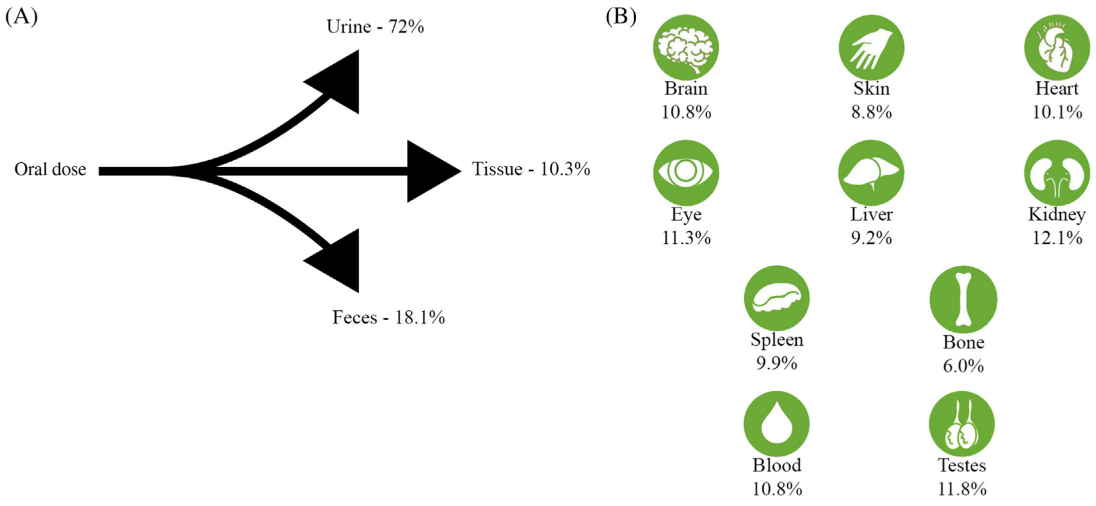

FIGURE 27.1 **Tissue distribution and bioavailability of MSM.** Elimination and tissue distribution of MSM in rats 48 h postoral ingestion. (A) Approximately 90% of orally ingested MSM is eliminated via urine or feces with approximately 10% remaining in the body 48 h after ingestion; percentages displayed from Otsuki et al.[3] data; (B) The remaining 10% appears to be roughly evenly distributed throughout tissues; percentages adapted from Magnuson et al.[4] data.

in humans is expected to be similar to those observed in rats as MSM has been shown to cross the blood−brain barrier (BBB) in humans as well.[6]

Currently, MSM is most commonly used as a treatment for arthritis[7,8] and interstitial cystitis.[9] However, with its broad bodily distribution and pleiotropic effects, MSM may have many other potential preventative applications. In the following sections, the potential cellular and tissue effects of MSM as an antioxidant will be discussed along with the possible influences on pathophysiology.

Overview of antioxidant effects

Biophysical properties

The unique physicochemical properties of MSM lend themselves to interesting biophysical effects, particularly on membrane interactions. To better understand the antioxidant functions of MSM, the biophysical effects of MSM and the more heavily researched DMSO should be discussed as these may influence the function of the double membrane−bound organelle and primary producer of reactive oxygen species (ROS), the mitochondria. Mitochondrial dysfunction is believed to be a major contributor to the pathogenesis of many disease states.[10]

With respect to membrane interactions, DMSO has previously been shown to modulate the structure and properties of cell membranes[11] and the adjacent water ordering[12] in a concentration-dependent manner. At low concentrations, DMSO causes the membrane to thin. As the concentration of DMSO increases, pores form as DMSO interacts with polar phospholipids. Beyond a concentration threshold around 25% molar concentration for DMSO, the membrane disintegrates in the hygroscopic medium,[11] while concentrations above 0.1 molar DMSO fraction result in dehydration of the lipid membranes and reduced intermembrane distances.[12] Together these biophysical effects may influence the electrochemical gradient and could have implications in vesicle secretion as these often involve a change in membrane potential or depolarization. Madji et al. reported that DMSO increases exocytotic neurotransmitter release.[13] Alternatively, these effects could also influence the mitochondrial membrane potential, and, in fact, DMSO has previously been shown to impair mitochondrial membrane integrity and alter the mitochondrial membrane potential in vitro.[14]

The mentioned biophysical effects of DMSO may be similar for MSM as both are commonly used organic solvents and effective co-transporters. Moreover, MSM may reduce the mitochondrial membrane potential of both cancerous[15] and noncancerous cells[16] through similar actions.

Antioxidant enzyme production/activity

ROS are essential components of cell signaling, but when produced in excess can damage lipids, proteins, and nucleic acids. That said, a homeostatic range of intracellular ROS must be maintained by balancing ROS production and both nonenzymatic and enzymatic antioxidants. Nuclear factor erythroid 2-related factor 2 (Nrf2) is emerging as a key modulator of oxidative stress by sensing redox status and regulating antioxidant-related gene expression. Upon activation of Nrf2 by ROS, Nrf2 translocates to the nucleus and upregulates superoxide dismutase (SOD)-3, peroxiredoxin, thioredoxin, thioredoxin reductase, and many glutathione-related genes.

MSM has previously been demonstrated to attenuate toxin-induced reductions in antioxidant enzymatic activities, specifically that of SOD[17–19] and catalase (CAT)[17–20] in multiple tissues. These alterations may be attributed to an increase in the translocation of Nrf2 to the nucleus, an effect previously observed in a neuroblastoma cell line.[21] Increased activities of SOD and CAT would improve the neutralization of superoxide radical (O_2^-) and hydrogen peroxide (H_2O_2) respectively, and possibly hydroxyl radical (OH^-) production from the dissociation of H_2O_2.

The diverse cellular effects of MSM can theoretically result in numerous changes in antioxidant enzyme production and activity. Presently, pretreatment with MSM has only been shown to attenuate toxin-induced reductions in SOD, CAT, and glutathione peroxidase (GPx)[19] as well as reduce myeloperoxidase (MPO) activity.[17,20,22] Though Nrf2 is known to upregulate glutathione-related genes, these effects do not always translate to changes in enzymatic activity. For example, increased translocation of Nrf2 to the nucleus does not change the activity of either GPx or glutathione S-transferase (GST).[21] If an increased production in glutathione synthesis genes is accomplished, an increase in intracellular glutathione concentrations could greatly enhance the redox hub activity, which will be discussed in the following section.

Redox hub

The redox hub is the inter-cycling of molecules between reduced and oxidized forms. This cycling can occur via enzymatic or nonenzymatic reactions. At the center of the hub is the cycling of reduced glutathione (GSH) and oxidized glutathione (GSSG). The interconversion of GSH and GSSG plays a central role as it can also affect the cycling of vitamin C between its reduced form, ascorbate, and its oxidized forms, monodehydroascorbate (mDHA) and dehydroascorbate (DHA) as well as convert lipid radicals to less reactive lipid hydroxides or hydroxy peroxide. Fig. 27.2 displays a simplified diagram of the redox hub with enzymatic and nonenzymatic reactions. In addition to regulating reduced and oxidized forms of certain molecules, the redox hub also contributes to the metabolic state of the cell as enzymes such as glutathione reductase (GR) oxidize reduced nicotinamide adenine dinucleotide phosphate (NADPH) to the oxidized form ($NADP^+$). These molecules are highly involved in multiple biosynthesis pathways.

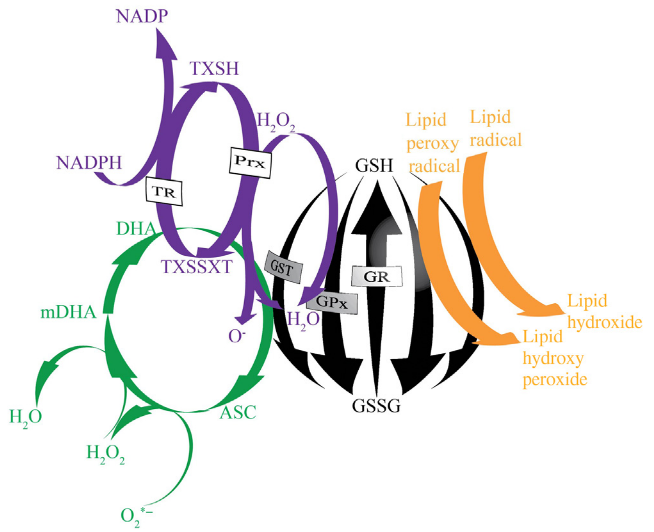

FIGURE 27.2 **Redox hub.** The redox hub represents intracellular small molecules that can take on either reduced or oxidized conformations. The oxidation/reduction of glutathione appears to interact with the reduction/oxidation of other intracellular small molecules. MSM may be able to influence the GSH:GSSG ratio.
ASC, Ascorbate; *H_2O*, water; *Prx*, peroxiredoxin; *TR*, thioredoxin reductase; *TXSH*, reduced thioredoxins; *TXSSXT*, oxidized thioredoxins.

The ratio of GSH/GSSG is often used as an indicator of oxidative status. In a previous in vitro study, MSM pretreatment reportedly restored the GSH/GSSG ratio following human immunodeficiency virus type 1 (HIV-1)-Tat exposure.[21] In terms of GSH, Tat exposure with MSM pretreatment was significantly greater than the levels from Tat exposure alone, while GSSG levels were significantly lower in Tat exposure with MSM versus Tat alone. Similar results have been found in a variety of tissues, such that GSH with MSM treatment was significantly increased[17,19,20,22–24] and GSSG levels were significantly decreased.[19] However, given the dynamic nature of these markers, these results are not always observed.[25–27]

Free radical scavenging

Free radical scavenging was a proposed antioxidant mechanism of MSM following a single cell–free experiment in which MSM showed the ability to reduce the quenching activity of hypochlorite on *O*-phenylenediamine (OPD) oxidation, though this effect was much less pronounced than with DMSO.[28] It is plausible that MSM may have been reduced to DMSO by OPD, thereby providing the quenching power as sulfoxides can then be oxidized to sulfones through reactions with hypochlorite. No other models of free radical scavenging have been proposed or tested. Therefore this is one of the less likely mechanisms of antioxidant function for MSM.

Methylsulfonylmethane in oxidative stress pathology

Inflammation

Chronic low-grade inflammation is a component of nearly all disease pathologies. Inflammation and oxidative stress have an interdependent relationship, whereby excess ROS can induce proinflammatory cytokine production and/or cytokine signaling can include ROS production. At the center of this interdependent relationship usually lie inflammasomes, multiprotein complexes responsible for the activation of the inflammatory response. The NLRP3 inflammasome has been linked to the pathogenesis of several diseases.[29] In short, a proinflammatory priming signal triggers NLRP3 monomer transcription and translation. Upon activation by one of several signals including ROS generation, NLRP3 monomers oligomerize to form the activated NLRP3 inflammasome, which can then promote the maturation and secretion of proinflammatory cytokines. Additionally, activated NLRP3 may also degrade the antioxidant enzyme regulator Nrf2.

MSM may modulate oxidative stress–inflammation crosstalk at the priming and/or activation step(s). During the priming step, MSM may reduce the activity of the proinflammatory NFκB transcription factor. This reduces the transcription of the NLRP3 monomer, thus, resulting in impaired NLRP3 inflammasome assembly and reduced suppression on Nrf2. Fig. 27.3 displays the potential actions of MSM, ultimately leading to a reduction in ROS available to act as a stimulatory signal for both NFκB nuclear translocation and NLRP3 inflammasome assembly.

Previous MSM studies using lipopolysaccharide (LPS)-induced toll-like receptor (TLR)-4 activation support this. For example, Kim et al.[30] reported that MSM reduced the nuclear translocation of NFκB. Furthermore, Ahn et al. reported that NLRP3 inflammasome activation was inhibited with less intracellular ROS.[31] By mediating this crosstalk between inflammation and oxidative stress, MSM may alleviate the chronic low-grade inflammation associated with the pathogenesis of many chronic disorders.

Brain

In addition to cognitive processing, the human brain indirectly coordinates numerous peripheral organ functions through the activation of various neuroendocrine axes such as the hypothalamic-pituitary-adrenal and the hypothalamic-pituitary-thyroid axes. These dynamic functions require the brain to constantly remain plastic. In order to handle all of these functions, the brain has an extremely high energy and oxygen demand. In fact, the brain consumes ∼20% of the body's total oxygen consumption to support this demand. This large oxidative metabolism makes the brain susceptible to the overproduction of ROS. Major brain sources of O_2^- are derived from NADPH oxidase (NOX) activity and complex I of the mitochondrial energy metabolism.[32] Within the brain, H_2O_2 is an important ROS/signaling molecule

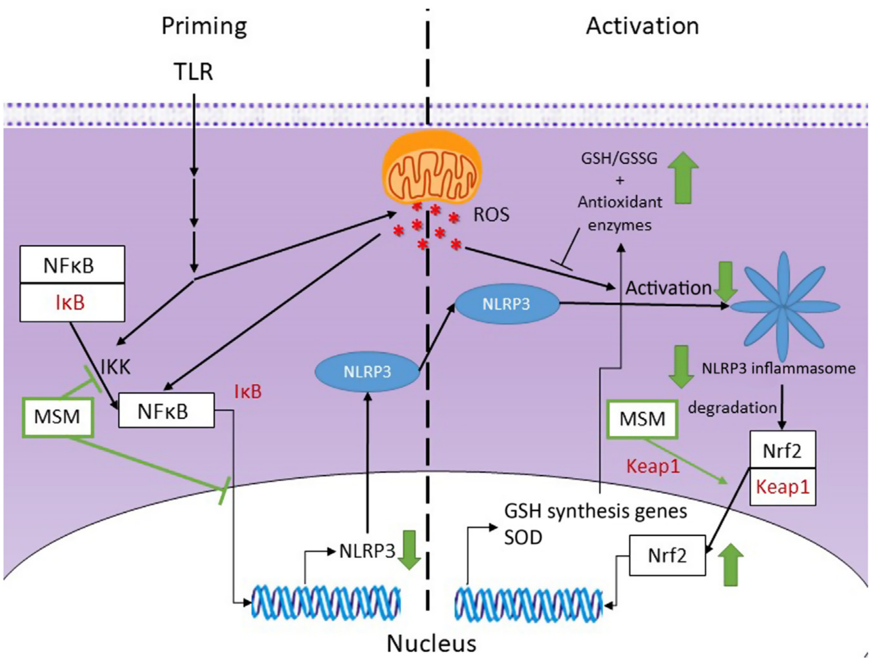

FIGURE 27.3 **MSM mediation of oxidative stress and inflammation.** MSM has been demonstrated to influence both the priming and activation steps of NLRP3 inflammasome assembly. Green arrows display the suggested net effect of MSM treatment.
IκB, nuclear factor of kappa light polypeptide gene enhancer in B-cells inhibitor; *Keap1*, Kelch-like ECH associated protein 1.

as it's critically involved in synaptic plasticity, but can also be catalyzed to the highly reactive OH^-. OH^- are very reactive with polyunsaturated fatty acids (PUFAs), substrates found in abundance within the brain. The thioredoxin/peroxiredoxin antioxidant system (Trx/Prx) of brain mitochondria appears to be a preferred method of H_2O_2 degradation rather than CAT.[32] Overwhelming of the brain antioxidant systems is implicated in the pathology of neurodegenerative disorders and traumatic brain injuries as ROS induce damage to cellular components such as PUFAs, thereby affecting membrane integrity and causing mitochondrial DNA damage, which can affect mitochondrial function.

MSM crosses the BBB where it is equally distributed between white and gray matter.[6] Previously MSM has been shown to decrease neuronal cell NO and ROS secretion while restoring the GSH/GSSG ratio upon dosing with HIV-1 Tat. This is believed to be mediated by the improved nuclear translocation of Nrf2[21] with MSM treatment. More recently, MSM has also been shown to reduce ROS and RNS following HIV-1 Tat exposure. This was accompanied by increased medial frontal cortex GSH levels and improved behavior in mice exposed to HIV-1 Tat.[33] Together, these studies suggest MSM can affect the cycling of the redox hub within the brain.

Cardiopulmonary

The cardiopulmonary system, inclusive of the heart, lungs, and vasculature, is tasked with extracting oxygen and circulating nutrients to all cells of the body while removing metabolic waste. To accomplish this, continuous, rhythmic contractions of the heart are required to pump blood to the lungs for oxygen diffusion and elsewhere. This constant requirement for energy by the cardiac muscle and exposure to ambient air and oxygen in the lungs puts this system at risk of the overproduction of ROS, in particular when oxygen demands are significant. In the heart, ROS from cardiac mitochondria, NOX2, and/or uncoupled endothelial nitric oxide synthase can target the sarcoplasmic reticulum or transverse tubule and disrupt calcium handling. Disruptions in calcium release can result in arrhythmia and if left untreated for an extended period of time can further lead to cardiac remodeling and eventually heart failure.[34] Within the vasculature, excess NO production from increased inducible nitric oxide synthase (iNOS) expression may result in impaired vasoconstriction. Because the lungs are readily exposed to environmental air, alveolar macrophage and goblet cell functions are important to prevent infections and produce mucus for gas diffusion respectively. An overproduction of ROS and lipid peroxide byproducts like isoprostanes within the lungs can affect nearby smooth muscle and provoke bronchoconstriction and vasoconstriction.[35] Within the heart and lungs, the most important antioxidant systems include mitochondrial CAT and the Trx/Prx and GSH/GSSG systems; while the most important feature in the vasculature is the regulation of iNOS expression. The overproduction of ROS is implicated in the pathogenesis of pulmonary hypertension, bronchopulmonary dysplasia, and many other cardiopulmonary disorders.

To date, few studies have investigated the effects of MSM on the heart. An in vitro study of cardiomyocytes indicated no oxidative stress related changes with MSM treatment,[26] however, MSM did exhibit an antiinflammatory effect in response to tumor necrosis factor-α (TNFα). In contrast, MSM may be an effective preventative treatment for lung injuries. MSM has previously been shown to protect against pulmonary hypertension by reducing mean arterial pressure and right ventricular systolic pressure while attenuating the monocrotoline-induced reductions of CAT, GPx, and SOD activities and GSH levels.[19] Backflow of blood is commonly seen with pulmonary hypertension and, thus, increases the right ventricular systolic pressure. If left untreated, this often causes heart failure. In another study, mice pretreated with MSM prior to paraquat-induced acute lung injury displayed increased SOD and CAT activities in the lungs along with improved oxidative stress markers such as increased GSH and decreased malondialdehyde (MDA), TNFα, and MPO activity.[17] Similar effects were not observed by DiSilvestro et al.[23] when mice were treated with carbon tetrachloride (CCl_4).

These results suggest MSM may be an effective indirect antioxidant within these tissues. MSM appears to increase the antioxidant enzyme activities of SOD, CAT, and GPx. These enzymes may also influence the cycling of the redox hub and affect GSH/GSSG ratios and Trx/Prx systems within these tissues, which could further promote protection against lipid radicals.

Kidney

In order to remove water soluble toxins and regulate the osmolarity and osmolality of the blood, the proximal tubules of glomeruli must generate enough energy to biosynthesize transporters and power ATP-requiring transporters. Thus, the cells comprising the proximal tubules are most susceptible to oxidative stress.[36] Oxidative stress, primarily from OH^- or peroxynitrite ($ONOO^-$) radicals, can target membrane lipids and impair membrane integrity and permeability, which can then lead to acute tubular necrosis. Resulting tubule damage ultimately reduces the glomerular filtration rate and can cause kidney injury. To mediate these damaging effects, the main antioxidants that the kidneys rely on are vitamins C and E, GSH, SOD, CAT, and GPx.[37]

MSM has been shown to be protective against glycerol-induced acute renal failure in rats.[38] This was accompanied by significantly lower blood urea and serum creatinine levels compared to those receiving glycerol alone; moreover, GSH levels were markedly increased after MSM treatment. Though much additional research is needed, results from this study suggest MSM may be an effective antioxidant by mediating the cycling of the redox hub within the kidneys.

Liver

The liver serves multiple metabolic functions from regulating glucose and lipid stores and secretion to degrading metabolic waste, drugs, and chemicals into excretable compounds. In addition to mitochondrial-derived ROS, cytochrome p450 enzymes located on the endoplasmic reticulum catalyze the degradation of toxins and produce O^{2-} as a byproduct. Because of the multifunctional role of hepatocytes and, thus, numerous potential sources of ROS, an overproduction of ROS is more likely than for other cell types. A change to the redox state of the hepatocyte often affects redox-sensitive transcription factors and can be responsible for altered protein expression within a stressed cell. Oxidative stress appears to be an important component in the pathogenesis of a number of liver-related injuries including nonalcoholic fatty liver disease, hepatic encephalopathy, hepatic fibrosis, and other liver diseases.[39]

MSM has been demonstrated to protect against acetaminophen-induced hepatotoxicity by attenuating SOD activity and GSH levels, while reducing MPO activity and MDA levels.[22] In another study, mice pretreated with MSM prior to paraquat-induced acute liver injury displayed increased SOD and CAT activities in the liver along with improved oxidative stress markers such as increased GSH and decreased MDA, TNFα, and MPO activity.[17] In another study, rats treated with MSM and CCl_4 showed reduced MDA, TNFα, and interleukin-6 (IL-6) levels and increased SOD and CAT activities.[18] Mitochondrial effects were also suggested through a reduction in Bax:Bcl2 ratio. Liver enzymes ALT, AST, and CYP2e1 expression were also increased. DiSilvestro et al.[23] demonstrated that mice treated with MSM and CCl4 produced a significant increase in GSH and this was accompanied by improved liver enzymes. These results suggest MSM may effectively modulate antioxidant enzyme activity and redox hub cycling within the liver.

Gastrointestinal tract

The gastrointestinal tract regularly encounters new pathogens and toxins that are often inadvertently ingested with food. As such, gastrointestinal (GI) tissue has many local immune cells that can contribute to the local ROS from epithelia encountering pathogens/toxins or from gut flora.[40] An overproduction of ROS within the epithelia may target tight junctions and affect the permeability of the GI tract and/or activate the local immune cells and contribute to inflammation.[41] Alternatively, the microbiome has been suggested to affect a number of different disease pathologies in the gut and other organs.[42]

Several MSM studies have demonstrated a protective effect of MSM on the GI tract. Two studies have utilized a model of acetic acid-induced colitis in rats.[20,43] Treatment with MSM daily for 4 days resulted in lower colonic levels of MDA and MPO activity, but increases in GSH and CAT activity. Both studies demonstrated significant decreases in inflammation, one using IL-1β, and the other through histological evaluation. Additional studies have shown the ability of MSM to protect against gastric mucosal injury from acidified ethanol and bisphosphonates.[44,45] Interestingly, MSM was previously shown to be one of the strongest serum biomarkers for patients with inflammatory bowel disease (IBD) based on nuclear magnetic resonance fingerprinting,[46] as IBD patients had far lower serum MSM concentrations than patients without IBD. This may suggest a lack of MSM-producing bacteria within the gut microbiome. Others have suggested microbiome sulfur metabolism may influence host-metabolism and oxidative stress.[47] These changes may be mediated by microbiome production of antiinflammatory short chain fatty acids.

Currently, MSM effects on the GI tract are limited, but show promise via changes to antioxidant enzyme activities and the cycling of the redox hub. Future studies may focus on the GI effects of MSM supplementation from a microbiome perspective as well as changes at the tissue level.

Musculoskeletal system and exercise

Physical inactivity is suggested to be a major contributor to chronic disease development.[48] As such, most healthcare providers often encourage physical exercise as a preventative measure despite the fact that strenuous physical exercise has the potential to generate ROS and lead to an acute state of oxidative stress. While this is the case, particularly when exercise is performed by those who are unaccustomed to such a stressor (i.e., untrained), the degree of oxidative stress is generally low and transient. Regardless, many individuals seek methods of lessening the oxidative burden and turn to dietary supplements to aid in this quest.

Evidence indicates that preloading with MSM prior to strenuous exercise can have promising results in blunting exercise-induced oxidative stress. Following aerobic exercise, MSM effectively reduces MDA,[27,49] protein oxidation,[27,49] and bilirubin[27,50] postexercise and increases total antioxidant capacity (TAC).[27,50] Glutathione levels are not as consistent; one study reported an increase to GSH and decrease in GSSG with MSM preloading and exercise,[49] while another reported no significant differences.[27] Similar discrepancies are also noted following anaerobic exercise, where one study reported an increase in Trolox equivalent antioxidant capacity (TEAC),[25] while another study found no differences.[25]

In addition to acute exercise, many individuals have used MSM to lessen the pain resulting from arthritis. In fact, arthritis is likely the most commonly cited reason for using MSM. Chronic low-grade systemic inflammation appears to drive the overproduction of ROS within the arthritic joint.[51] By doing so, degeneration of the cartilage and synovium within the joint space contribute to the pathology of arthritis. This arthritic pain is often a hindrance to individuals trying to partake in physical activity[52,53] and evidence supports the use of MSM to provide relief from the pain of arthritis.[7,8]

Applications in other areas of pathology

MSM may be able to indirectly influence ROS production by improving glucose homeostasis. In mouse models of diet-induced obesity and genetically obese diabetic animals, MSM treatment reduced blood glucose, hepatic triglyceride, and cholesterol levels and made mice hypersensitive to insulin.[54] Improvements to measures such as these may indirectly provide protection from the overproduction of ROS as many of these variables are known to stimulate mitochondrial activity and, thus, ROS generation.

Interestingly, much research has focused on the anticancer effects of MSM. Caron and colleagues have suggested that MSM treatment of metastatic cancer cells is able to revert the metastatic phenotype.[55–58] Other studies have shown MSM can inhibit cancer cell viability at concentrations of approximately 300 mM.

Because MSM shows broad tissue distribution, it may be able to alleviate tissue injuries at many levels. It is important to keep in mind that each of these tissues can greatly affect other tissues, either positively or negatively. For instance, liver failure can cause renal failure, which can cause metabolic acidosis resulting in oxidative stress within the brain and encephalopathy. A generalized overview of these interacting pathways is provided in Fig. 27.4.

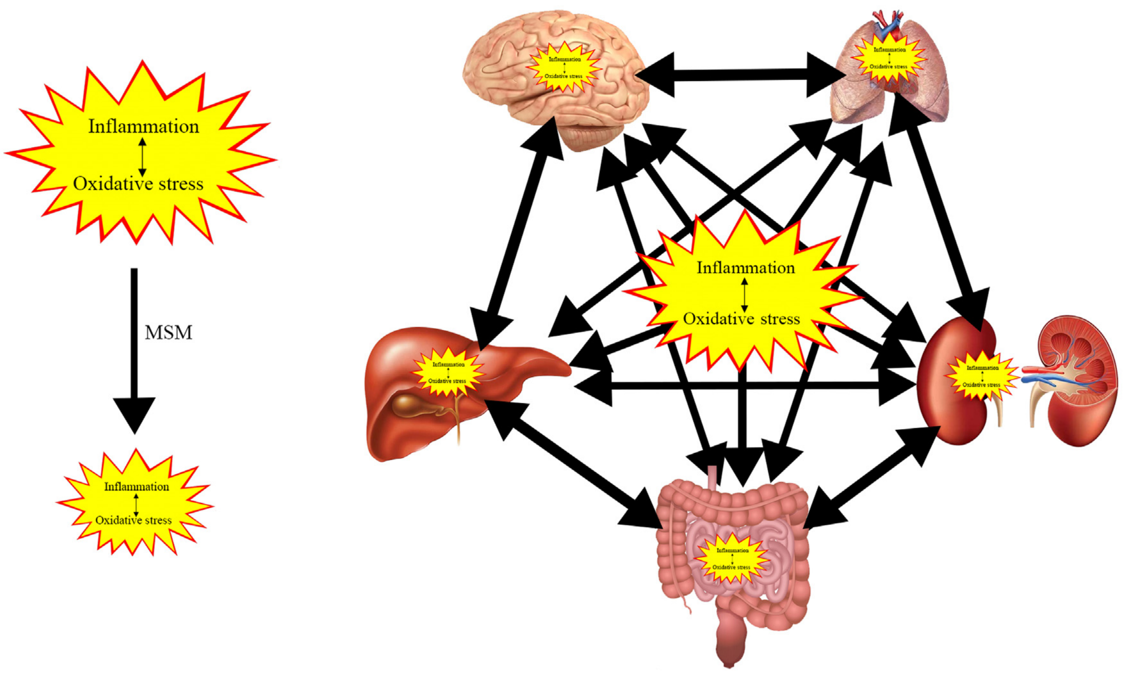

FIGURE 27.4 **Simplistic scheme of interorgan pathologies.** Because MSM can penetrate multiple tissues, it may influence the pathophysiological state of one tissue, which can then affect the pathological state of other tissues. This interrelationship may span throughout the entire organism making defining an underlying mechanism nearly impossible.

TABLE 27.2 Overview of oxidative stress markers effected by MSM.

Study type	Stimulus/tissue	Result		Reference
Human	Exercise-induced oxidative stress	MDA	MSM blunted the postexercise rise in MDA 2 and 24 h postexercise	49
		GSH/GSSG	Increase in plasma GSH and decrease in GSSG in MSM group postexercise; GSH/GSSG ratio preexercise, immediately postexercise, 30 min, and 2 h postexercise	
		PC	Significant decrease postexercise	
Human	Exercise-induced oxidative stress	MDA	Pretreatment with oral MSM resulted in reduced MDA levels preexercise and 2 h postexercise despite not inducing overall increase	27
		GSH/GSSG	No differences	
		TAC	Maintained TAC elevation 24 h postexercise	
		PC	Significant decrease 2 and 24 h postexercise	
		Bilirubin	Significantly lower immediately after exercise	
		Uric acid	Significantly reduced serum uric acid levels postexercise	
Human	Exercise-induced oxidative stress	MDA	No change	59
Human	Osteoarthritis	MDA	MSM group significantly lowered levels of MDA in urine at 12 weeks versus placebo	8
Human	Exercise-induced oxidative stress	TAC	Significant increase in TAC 2 and 24 h after exercise	50
		Bilirubin	Significantly lower 2 and 24 h after exercise	
Human	Exercise-induced oxidative stress	GSH/GSSG	No differences	25
		TEAC	TEAC increased significantly following exercise	
Human	Exercise-induced oxidative stress	SOD	No change	60
		TEAC	No change	
Animal	HIV-1 Tat	GSH/GSSG	Increased GSH towards baseline; reduced GSSG towards baseline; increased GSH/GSSG	21
		ROS	Reduced ROS/RNS	
Animal	HIV-1 Tat	ROS	Reduced ROS/RNS	33
Animal	Glycerol-induced renal failure	GSH/GSSG	Attenuated reduction in GSH	38
		Urea	Significantly reduced	
Animal	Acetate-induced colitis	MDA	Pretreatment with oral MSM attenuated rise in MDA	20
		GSH/GSSG	Increased GSH levels of colonic tissues; prevented colonic depletion of GSH	
		CAT	Elevation in CAT activity	
		MPO	Decrease in MPO activity	
Animal	Paraquat-induced lung and liver injury	MDA	Pretreatment with oral MSM attenuated rise in MDA	17
		GSH/GSSG	Attenuated GSH loss in liver; pretreatment prevented the reduction of hepatic GSH content by paraquat	
		CAT	Elevation in CAT activity	
		SOD	Elevation in SOD activity	
		MPO	MPO activity decreased in lung and liver	
Animal	Monocrotoline-induced pulmonary hypertension	MDA	Pretreatment with oral MSM attenuated rise in MDA dose-dependently (maximal suppression of 60% in response to 400 mg/kg/day)	19

(*Continued*)

TABLE 27.2 (Continued)

Study type	Stimulus/tissue	Result		Reference
		GSH/GSSG	Significantly restored levels of GSH and GSSG. Value of GSH/GSSG ratio significantly lower in PAH, but recovered to control levels after the treatment and further enhanced at a higher dose	
		CAT	Elevation in CAT activity	
		SOD	Elevation in SOD activity	
		GPx	Dose-proportional increase in GPX activity	
Animal	CCl4-induced acute liver injury	MDA	Pretreatment with MSM attenuated rise in MDA	18
		CAT	Elevation in CAT activity	
		SOD	Elevation in SOD activity	
Animal	Acetaminophen-induced hepatotoxicity	MDA	Pretreatment with MSM attenuated rise in MDA	22
		GSH/GSSG	Pretreatment with MSM led to GSH reduction in hepatic tissue	
		SOD	Elevation in SOD activity	
		MPO	Attenuation of MPO activity	
Animal	Exercise-induced oxidative stress	Lipid radicals	Reduced levels of lipid peroxidases	24
		GSH/GSSG	Increase in GSH; result potentiated by vitamin C	
		NO	Reduced NO plasma levels after several weeks of intense exercise	
		CO	Reduction in CO release	
Animal	CCl4-induced acute liver injury	GSH/GSSG	Increased GSH in liver; no change in lung or skeletal muscle	23
In vitro	LPS-induced inflammation of macrophages	ROS	Significant reduction	31
In vitro	TNF-α-induced inflammation of cardiomyocytes	GSH/GSSG	No differences	26
		TAC	No difference	

A summary of peer-reviewed studies including antioxidant markers is shown above.

Conclusion

The unique properties of MSM allow for broad tissue distribution throughout the body. By affecting the redox status of one or several systems, MSM can potentially influence the pathogenesis of numerous disorders. MSM likely provides protection from tissue injury by mediating the crosstalk between oxidative stress and inflammation. Additional research is needed to determine the entirety of therapeutic benefits owing to MSM (Table 27.2).

Summary points

- MSM is naturally occurring, commonly found in onions and garlic, and nontoxic.
- MSM is widely distributed throughout bodily tissues.
- MSM alters activities of superoxide dismutase, catalase, and myeloperoxidase antioxidant enzymes in a variety of tissues.
- MSM appears to affect the cycling of reduced and oxidized glutathione, ultimately affecting the redox hub and redox status.
- In addition to antioxidant activity, MSM also demonstrates an antiinflammatory effect suggesting it can mediate the crosstalk between these processes.

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